

Biological aging clocks in health and disease

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 Check for updatesTony Wyss-Coray^{1,2,3}✉ & Eric J. Topol⁴✉

There are extensive ongoing efforts to slow or even reverse human aging, such as with epigenetic cellular reprogramming, thymus rejuvenation or senolytics. In parallel, new and diverse metrics—known as biological clocks—have been discovered and shown to track the pace of aging in an individual and their organs, tissues and cells. These clocks have multiple potential use cases, including identifying people at high risk of disease, serving as a foundation for prevention or early detection, and determining whether lifestyle factors or an intervention can modulate the aging process. This review provides a critical appraisal of the progress that is being made with biological clocks and how they might ultimately help understand pathobiology, reduce the burden of disease and extend healthspan.

Across cultures, humans have long sought to extend lifespan and delay aging. Once viewed as speculative, rejuvenation and longevity are now mainstream scientific pursuits, enabled by converging advances in molecular biology, genomics, high-throughput proteomics, single-cell technologies and artificial intelligence (AI). Average human lifespan has nearly doubled over the past century, but healthspan, the years lived in good health, has not increased at the same rate. Consequently, more people across the globe are with age-related cardiovascular disease, cancer, dementia and frailty.

At the same time, aging research advanced to demonstrate that aging is not a single, uniform process but a collection of overlapping molecular and physiological changes that unfold, at different rates, within an individual. One of the most important advances is the ability to use biological clocks to quantify the aging process at the whole organism, organ and cellular levels. From the first epigenetic clocks built on DNA methylation patterns to plasma proteomic clocks for individual organs and cells, AI-based readouts of the retina and emerging multi-omic and electronic health record (EHR)-based metrics, the field now possesses a sizeable toolkit for tracking how each person ages. These research tools reveal nonlinear and asynchronous trajectories of aging and functional decline, linking lifestyle and exposures to molecular changes. They have the potential to identify individuals at elevated risk for specific diseases years before symptoms, and to serve as early endpoints for trials of geroprotective interventions. While there are multiple caveats and a lack of readiness for use in medical practice, molecular clocks lay the foundation for tracking the aging process for people in the future. This could ultimately be used to help

slow the aging process and improve healthspan, be it through lifestyle changes, drug interventions or other approaches.

In this review, we discuss the rapidly evolving landscape of biological aging clocks and their emerging role in health and disease. First, we summarize key conceptual advances in the molecular biology of aging that have laid the foundations for aging clocks. We then review the major classes of aging clocks from epigenetic, proteomic, cellular, immune or imaging based, to integrative multi-omic metrics. We discuss how these clocks could move from research into the clinic, including their potential to support preventive medicine, surrogate endpoints in trials of geroprotective drugs and individualize surveillance frameworks for diseases such as Alzheimer's disease (AD). Finally, we consider the technical, regulatory and ethical challenges that must be addressed before aging clocks can become a routine part of medical care.

Progress in the science of aging

Aging as a molecular process

From its historical roots in metaphysical and philosophical speculation to the emergence of aging theories in the 20th century, aging research has made major advances, converging on a molecular and cellular understanding of its underlying mechanisms. This progress has been fueled by the study of biological outliers with genetic mutations or exceptional adaptations to their environment, and by the application of the Darwinian descriptive method combined with modern, unbiased quantitative measurements ranging from imaging to omics technologies. Early observations of extended lifespan in dwarf mouse strains,

¹Department of Neurology and Neurological Sciences, Stanford University School of Medicine, Stanford, CA, USA. ²The Phil and Penny Knight Initiative for Brain Resilience, Stanford University, Stanford, CA, USA. ³Wu Tsai Neurosciences Institute, Stanford University, Stanford, CA, USA. ⁴Scripps Research Translational Institute, Scripps Research, La Jolla, CA, USA. ✉e-mail: twc@stanford.edu; etopol@scripps.edu

linked to reduced growth hormone production, were mechanistically explained by genetic studies in *Caenorhabditis elegans*¹. These worm studies revealed that mutations in the gene encoding *DAF-2*, a homolog of insulin growth factor-1 (which is regulated by growth hormone), could double normal lifespan. Further work identified the transcription factor *DAF-16* (a homolog of *FOXO*) as key regulator of insulin/IGF-1 signaling¹. Human *FOXO* genes are associated with longevity and *IGF1* variants with body size and lifespan². Beyond nutrient sensing, variants in *LMNA*, *WRN*, *ERRC1* and *ERCC4* (also known as *XPF*) linked to accelerated aging and progeroid syndromes in humans highlight the roles for DNA repair, nuclear architecture and chromatin stability. Similarly, centenarian studies connect *APOE* ϵ 2 variants and loss-of-function CETP variants to lipid metabolism and longevity³. Collectively, these findings suggest that lifespan is shaped by reduced damage accumulation across genome maintenance, nutrient sensing, inflammation and stress resistance³.

The technological advances of the twenty-first century have dramatically expanded our capacity to delineate molecular changes across the lifespan. Epigenetic changes demonstrate that age-related marks robustly track lifespan in an evolutionarily conserved way. Concurrent atlases of cellular and tissue-specific molecular changes—transcriptomes, proteomes and other modalities—now define shared and distinct temporal trajectories across cells, tissues and species. For example, a gene expression analysis in 20 mouse organs and tissues (Tabula Muris Senis) uncovered tissue-specific aging trajectories, circadian gene changes and accumulation of plasma B cells⁴. More recently, large-scale single-cell analysis across >300 cell types revealed patterns of immune cell dynamics with age⁵. Parallel efforts profiling the human proteome in 13 tissues over five decades identified broad shifts partially reflected in plasma proteomic clocks⁶. These datasets, increasingly integrated with AI modeling, are becoming more robust and hypothesis-generating, pointing to new therapeutic targets.

Extensive modeling of genetic pathways has led to the formulation of ‘hallmarks’ of aging and, more recently, hallmarks of health^{2,7}. Key hallmarks of aging include genomic instability, telomere attrition, epigenetic alterations, loss of proteostasis, deregulated nutrient sensing, mitochondrial dysfunction, cellular senescence, stem cell exhaustion and altered intercellular communication. Conversely, hallmarks of health emphasize barrier integrity, homeostatic maintenance and stress resilience. This framework underscores the complexity of aging, suggesting that no single pathway provides a singular route to rejuvenation. Indeed, interventions such as exercise, caloric restriction and parabiosis engage overlapping and distinct molecular pathways⁸. Notably, unbiased atlas and phenotyping efforts, especially when paired with interventions that modify aging trajectories, are increasingly useful for prioritizing key mechanisms of aging across species.

Aging as a nonlinear and asynchronous physiological process

A major advance in our understanding of aging comes from the realization that, within an individual organism, cells, tissues and organs can age at different rates. Longitudinal observations in cohorts such as the Baltimore longitudinal study of aging demonstrated that individuals age at different rates, as do organs within the same individual⁹, supporting early studies in *C. elegans* that showed that age-related structural decline is tissue-specific and stochastic¹⁰. Quantifying DNA methylation across the lifespan demonstrates that aging is asynchronous across tissues and species, with distinct biological ages co-occurring within the same organism¹¹. Advances in RNA sequencing (RNA-seq) have enabled characterization of pronounced and tissue-specific temporal transcriptional changes with age^{4,12}. These observations were further explored using a technique known as heterochronic parabiosis, whereby mice of different ages are joined surgically, leading to a shared circulatory system. Single-cell gene expression and DNA methylation across 20 tissues in the mouse demonstrated that cells of young mice paired with old mice showed various degrees of ‘accelerated aging’, while cells of

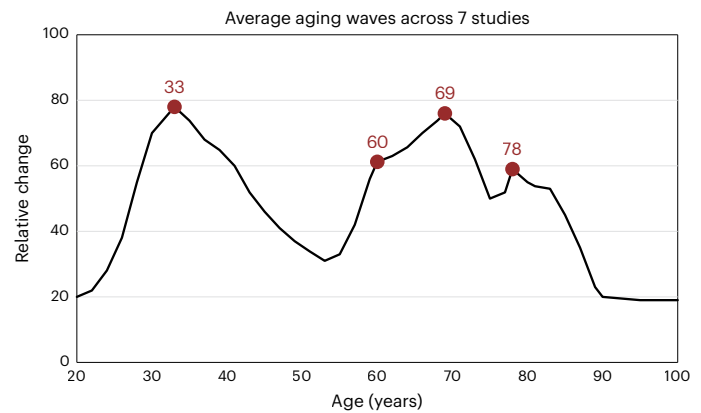


Fig. 1 | Nonlinearity of aging. The graph shows a consensus smoothed mean curve across seven studies, showing peaks at early life (age = 29, 32, 34 years), first late-life cluster (age = 57, 60, 60 years), second late-life cluster (age = 66, 67, 70, 72 years) and late life (age = 78, 78, 83 years). Data are from refs. 16,18–22.

old mice showed rejuvenation, indicating that, within the same organism, cells and tissues can have dissimilar molecular ages that are also different from the chronological age of the mouse^{13–15}.

The nonlinearity of aging at the population level was described in the blood proteome of more than 4,300 individuals aged 20 to 90 years, revealing coordinated proteomic shifts around 34, 60 and 78 years of age¹⁶. Epigenomic analyses of blood cells from healthy adults aged 22–93 years identified inflection points around ages 38–42 and 62–71 years, with modest sex differences¹⁷. Further support comes from longitudinal multi-omic profiling of 108 individuals, showing inflection points across lipids, metabolites, proteins, immune cells and microbiomes around 44 and 60 years¹⁸. Brain studies similarly reveal nonlinear aging trajectories. Plasma proteins associated with a composite aging score based on multimodal brain imaging showed coordinated shifts at ages 57, 70 and 78 years¹⁹. Functional magnetic resonance imaging (MRI) analyses show transitions in connectivity at ~40 and 70–75 years²⁰. Diffusion imaging across >4,000 brains identified topological turning points at ages 9, 32, 66 and 83 years, closely aligning with proteomic waves²¹. Large-scale structural imaging of >100,000 individuals further identified peaks in gray and white matter at ~7 and 29 years, followed by accelerated white matter loss beyond the age of 50 years²².

Collectively, these ‘waves’ of increased activity represent coordinated molecular and structural reprogramming across the lifespan. They point to three major phases of adult human aging (~30 to 40, 60 to 70 and ~80 years), marking transitions in systemic physiology and potential windows for intervention (Fig. 1).

Mutability of aging

A central question remains whether aging can be paused, slowed or even reversed. In principle, the answer is yes—during reproduction, germ cells fuse to form a zygote of biological age zero, effectively resetting hallmarks of aging. This feat was recapitulated in 2006–2007 through induction of pluripotency—through activation of four transcription factors known thereafter as the Yamanaka factors—in somatic cells^{23,24}. These induced pluripotent stem cells can generate any cell type and represent a major advance in regenerative biology. Epigenetic reprogramming thus enables the generation of donor-specific ‘young’ cells and tissues, and raises the possibility of organismal rejuvenation. Clinical efforts are now testing induced pluripotent stem cell-derived cells in patients with diseases including macular degeneration, heart disease and Parkinson’s disease^{25–27}. While aimed at repair, these approaches essentially replace aged tissues with young cells.

Beyond cell replacement, transient epigenetic reprogramming is attempting to rejuvenate tissues *in vivo*²⁸. Transient expression

Table 1 | Molecular clocks to track aging

Clock type	Models	Primary data input	Training target	Clinical utility	Key limitations
First generation	Horvath1,2	DNA methylation	Chronological age	Baseline; cancer	Lack of pathology
Second generation	PhenoAge, GrimAge, GrimAge2	DNA methylation, plasma proteins	Mortality, biological age	Predicts lifespan, healthspan, frailty	Plasma protein heavy influence
Third generation	DunedinPACE, DunedinPoAm	DNA methylation	Rate of physiological decline	Speed of aging, mortality risk	Requires serial assessment
Fourth generation (deep learning)	AdaptAge, CpGPT, SITH	DNA methylation, large-scale arrays	Chronological age	Precision age prediction	Lack of explainability
Organ clocks	Proteomic biological age gaps	Organ-derived plasma proteins	Chronological age	Specific disease prediction	Assay expense
Cell clocks	Proteomic biological age gaps	Cell-derived plasma proteins	Chronological age	Specific disease prediction	Assay expense

of reprogramming factors can reset cellular age without full dedifferentiation into stem cells. The cyclic activation of the Yamanaka factors reversed aging hallmarks and extended lifespan in a progeria mouse model, and improved tissue function in wild-type mice^{29,30}. Similarly, transient activation of these reprogramming factors in human fibroblasts reduced cellular age by ~30 years, without changing their identity³¹.

Systemic factors also modulate aging. Parabiosis experiments demonstrated that exposure to a young circulatory environment rejuvenated aged muscle stem cells³². In addition, while organs of aged mice were broadly rejuvenated at molecular, cellular and functional levels, aging was accelerated in the young counterparts^{13,15,33}. Plasma exchange studies in mice recapitulated these effects, leading to clinical studies in neurodegeneration and aging^{33–36}. Therapeutic plasma exchange has shown modest benefits in AD and reversed aging biomarkers in individuals over 50 years of age, indicating that biological age may be malleable in nondiseased, older adults^{37,38}.

Lifestyle interventions, including caloric restriction and exercise, as well as drugs such as rapamycin (an mTOR inhibitor), metformin (antihyperglycemic), glucagon-like peptide-1 (GLP-1) receptor agonists, the coenzyme NAD and omega-3 fatty acids, have also shown rejuvenating effects in animal studies³⁹, although the anti-aging effects of metformin have subsequently been questioned⁴⁰. One large, randomized study of more than 3,000 healthy adults aged 70 years or older studied the effects of exercise, omega-3 fatty acids and vitamin D on aging—reporting slowing of epigenetic aging in a post hoc analysis of 777 participants with omega-3 alone, and additive effects with vitamin D and exercise⁴¹. However, most other interventions remain insufficiently validated in healthy humans^{42,43}. Clinical translation of in vivo findings will require large, randomized trials and robust biomarkers of biological age to assess efficacy.

Clinical studies of clocks

To capture changes in physiological function across lifespan, computational models have been developed to estimate the biological, rather than chronological age of organisms, organs and cells. These biomarkers of aging, often referred to as clocks, capture individual physiological differences relative to a larger reference population. Their development has been accelerated by the work of several large consortia, including the Biomarkers of Aging Consortium and the Aging Biomarker Consortium^{44,45}. Table 1 summarizes key features of epigenetic (first to fourth generation) and proteomic clocks, and their progression over time. Of note, the fourth generation of DNA methylation clocks introduces deep learning transformer models to gain insights about causality, such as with AdaptAge, DamAge and Skip-Improved Training Hive (SITH) models, the latter arising from an open competition for biomarkers of aging^{46,47}.

Below, we provide an overview of recent advances in aging clocks based on epigenetic and other data, with a particular focus

on proteomic-based organ and cellular biomarkers as emerging and promising fields. These newer clocks have the potential to pinpoint, within an individual, an organ or cell type that shows accelerated aging.

Epigenetic clocks

An abundance of clinical studies has used molecular clocks for predicting various endpoints, including lifespan, healthspan, frailty and the pace of aging for whole body and organs, and specific medical conditions. The field got its start with the initial DNA methylation clock to predict chronological age in 2013 (and later the Horvath2 clock), which led to many other epigenetic-based clocks, particularly PhenoAge, GrimAge2 and DunedinPACE, which were widely assessed and used as benchmarks for accuracy of prediction of all-cause mortality and other major health outcomes^{11,48–51}.

An early study of centenarians demonstrated that their epigenetic age, according to the Horvath clock, was 8.6 years less than their chronological age⁵². Across 348 mammalian species, DNA methylation predicted maximal lifespan^{53,54}. Human epigenetic clocks vary in the first two decades of life, suggesting that what makes these clocks ‘tick’ in part originates in early development and is, in part, genetically controlled⁵⁵.

By adding clinical, laboratory and lifestyle-indexed CpG sites, PhenoAge predicts all-cause and cause-specific mortality, healthspan and chronological age in 35 different tissue and cell types⁴⁹. GrimAge2, a refinement of the first version (GrimAge) with more than fivefold participants and extensive longitudinal follow-up, provides prediction for lifespan, healthspan, time-to-coronary heart disease, cancer and other conditions⁵⁶. In a systematic review, GrimAge2 showed the best predictive accuracy for frailty⁵⁷. DunedinPACE used DNA methylation along with extensive clinical and lab data from a 1972–1973 birth cohort to derive a pace of aging metric and was replicable—with partitioning to slow, average or fast pace of body-wide aging—for survival outcome in other cohorts. There are many other clinical clock studies based on epigenetic markers, including Zhang 10, Hannum, DunedinPoAm, YingAdaptGe, YingCausAge, YingDamAge, DNAmTL, PathwayAge and several others⁵⁸. A methylation-based ‘intrinsic capacity’ clock—trained on indicators of functional and mental capacity, including hand strength, cognitive function, locomotion, eye and hearing acuity, and psychological testing—was found to correlate well with immune system function and inflammation^{59,60}. A Mendelian randomization analysis of four cohorts showed that the interferon pathway, with its capacity to induce low-grade chronic inflammation (inflammaging), was a driver of epigenetic age (according to Horvath, Hannum, GrimAge, PhenoAge)⁶¹. For endpoints of lifespan, healthspan, cancer, stroke and dementia, the GrimAge2, Horvath2 and DunedinPACE clocks were generally associated with the highest predictive capacity⁶².

The value of methylation clocks in capturing the risk of cardiovascular disease that was not discernible through established risk factors was shown in a retrospective study of more than 1,200

participants followed for 8 years⁶³. A study of 14 epigenetic clocks for prediction of 174 diseases found particularly strong links for lung and liver conditions⁵⁸.

Methylation clocks have specifically been assessed for their correlation with brain aging, as reviewed in ref. 64. Their association extends to cognitive performance, brain MRI results and risk of dementia.

Recent studies have identified interventions that appear to slow epigenetic aging. For example, the Shingles vaccine, which correlated with a lower incidence of AD and dementia in four large natural experiments, was shown to slow epigenetic aging in a large US cohort^{65,66}. Likewise, a daily multivitamin slowed epigenetic aging in a randomized trial⁶⁷. However, the effect size for both the vaccine and multivitamin was small, and it is unclear whether this would be linked to a meaningful change in lifespan or healthspan.

Instead of cross-sectional studies when only one measurement of epigenetic age is obtained, longitudinal assessments are expected to be more informative. As pointed out in ref. 68, there is considerable individual variability, highlighting the need for age and sex-stratification and longitudinal studies in diverse populations. While not diverse, a cohort of 699 adults in Italy followed for 24 years, with two measurements for 376 participants, and three measurements for 323 participants, found higher risk of death in people with faster increase in multiple clocks, independent of the baseline epigenetic age⁶⁹.

Overall, epigenetic clocks are validated metrics of the biology of aging and statistically correlate with various health outcomes. However, proving causality with these clocks would be technically challenging⁷⁰. Notably, their use at the individual level for clinical prediction has not been established. Whether they can be used to forecast clinically relevant outcomes, especially as a single measurement or as a single clock, is particularly uncertain. More longitudinal assessment in diverse populations with attention to attrition, survival bias and appropriate demographic stratification will help determine whether these measurements will become clinically relevant.

Proteomic clocks

A revolution of high-throughput proteomics has recently begun⁷¹. From as little as 50 µl of blood, up to 11,000 plasma proteins can be measured. While not a comprehensive assessment of the 20,000 protein-coding genes or more than 100,000 different proteins in our blood (arising as a result of alternative splicing), proteomics has been quite informative about health outcomes. Fortunately, we have multiple cohorts and studies to draw from, including some with longitudinal assessment. Together, these demonstrate that clusters of plasma proteins that tag the aging process of organs and cells (and the immune system) correlate with organ- and cell-specific health outcomes over extended follow-up. Furthermore, many interventions, such as lifestyle and medication, have been assessed for their impact on a proteomic-based organ clock. Below, we review the principal findings of key studies.

Organ clocks. An earlier study followed five cohorts (three with healthy participants and two with AD, together including nearly 6,000 adults) for 15 years and used machine learning and proteomic analysis to measure organ-specific aging⁷². The gap between the proteomic-based age of each organ and the individual's chronological age defined the pace of aging of each organ, and accelerated organ aging was associated with an increase in mortality. Moreover, the age of specific organs was linked to organ-associated disease (for example, heart failure linked to accelerated heart age and AD with brain age).

A similar study of 53,000 UK Biobank participants with nearly 3,000 plasma proteins identified seven organ clocks and confirmed the relationship of organ aging to organ-specific disease, such as liver age and cirrhosis, lung aging and chronic obstructive pulmonary disease⁷³. It also extended the relationship of organ clocks with lifestyle factors such as smoking, alcohol intake, specific foods and medications.

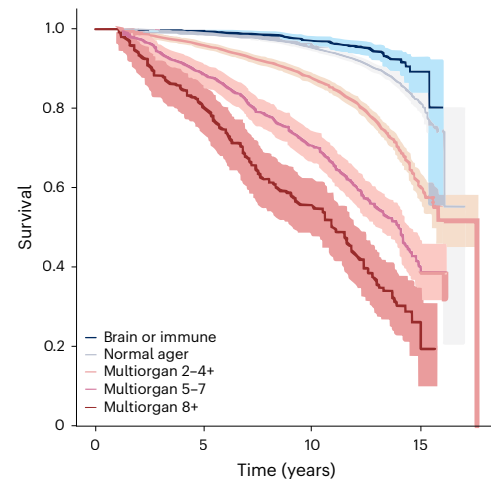


Fig. 2 | Accrual of aged organs progressively increases mortality risk, whereas brain and immune system youth is associated with longevity. Kaplan-Meier curves with 95% confidence intervals showing survival over a 17-year follow-up for normal agers, multi-organ agers (with 2–4, 5–7 or 8+ years aged organs) and individuals with a youthful brain or immune system (brain- or immune-) from the UK Biobank. Biological organ ages were calculated using linear regression modeling based on proteomic measurements in blood plasma obtained at time zero from individuals. Image is adapted from ref. 74; Springer Nature.

A study of 55,000 UK Biobank participants, with about 3,000 proteins analyzed across 11 organs over 17 years of follow-up, further confirmed prior findings—slowing associations between organ age and specific lifestyle factors and medications⁷⁴. A notable additional finding was that advanced proteomic-based brain age was associated with a 3.1-fold increase in the hazard of developing AD, independent of *APOE* genotype. The brain and immune system clocks were most prominently linked to survival over extended follow-up (Fig. 2). Postmenopausal estrogen was associated with slower aging across multiple organs, especially the immune system.

Organ-specific aging signatures were also identified in a prospective study of 6,235 participants, healthy at baseline, with a mean follow-up of nearly 20 years, further adding to the correlation between organ-specific aging and 30 diseases⁷⁵. Furthermore, the aging of particular organs did not show substantial correlation with the pace of aging of other organs, but individuals with fast-paced organ aging had heightened risk for 30 of 45 age-related diseases. The highest risk of dementia was linked to the pace of aging of the immune system rather than the brain. In a separate fifth study, this group also showed the association of proteomics with social disadvantage and socioeconomic status⁷⁶, highlighting the role of environmental factors in modulating the aging process.

Yet another multi-organ proteomic-based clock study of the almost 44,000 UK Biobank participants used genomic and EHR data to explore causal relationships between organ age gaps and disease outcomes⁷⁷. It remains unclear whether the proteins are the drivers of the aging process or the result. Future studies with molecular clocks may help deconvolute cause and effect. Mendelian randomization is a method traditionally used for this purpose. To that end, proteomic profiling of 4,775 proteins in ~8,000 healthy individuals with integration of genomic data with Mendelian randomization led to assertion of multiple causal protein–disease links⁷⁸. Likewise, integration of proteomic data (based on the next-generation Olink platform) with genomic data from 53,000 UK Biobank participants suggested several potential causal protein–disease relationships⁷⁹. Beyond that, many specific protein biomarker candidates were identified for tracking the process of biological aging⁸⁰.

Adding to these multiple cohorts and analyses is one of the most important studies on proteomic organ clocks conducted on the world's oldest, continuously followed birth cohort, the UK's National Survey

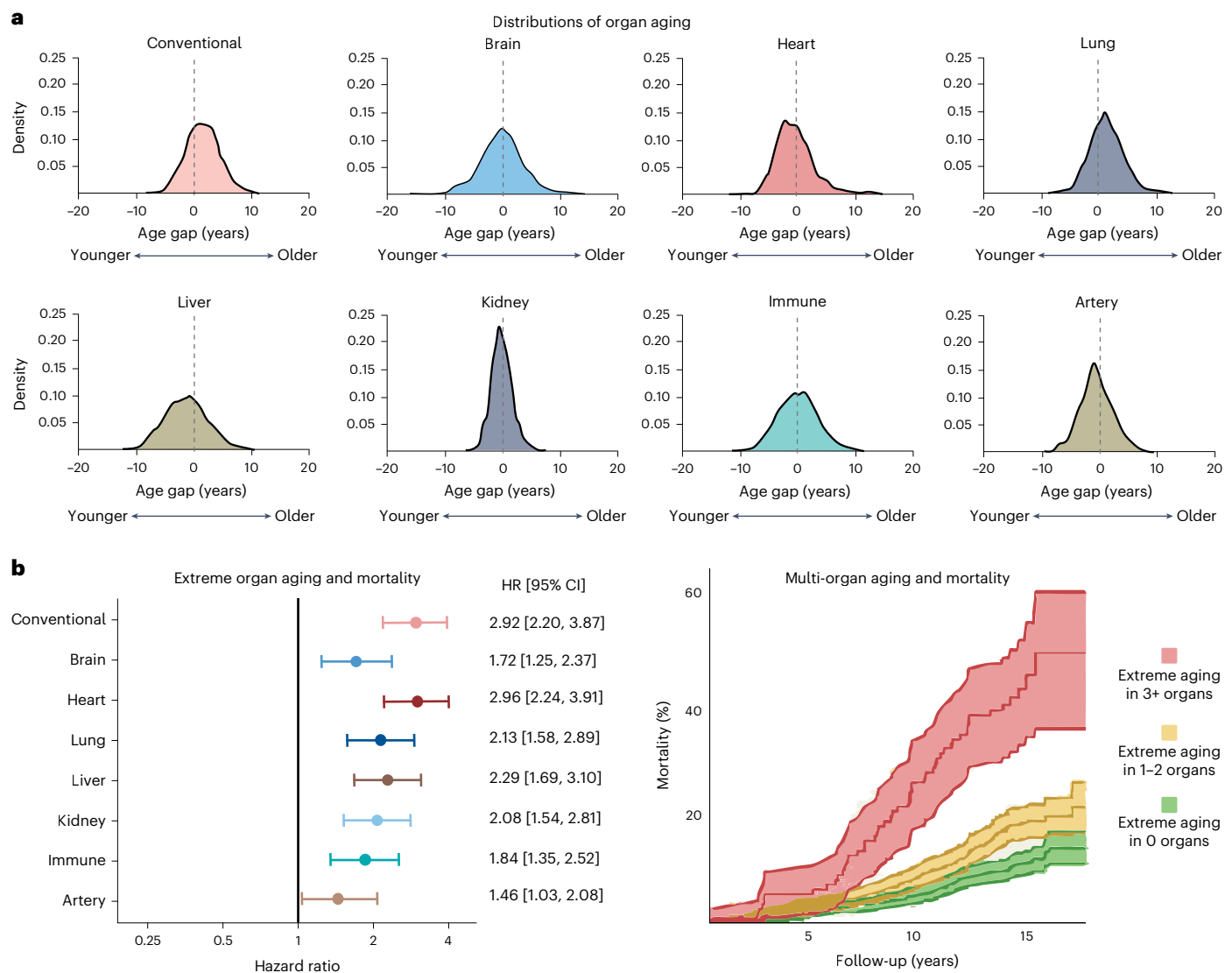


Fig. 3 | Organ aging heterogeneity in a same-aged birth cohort is prognostic for mortality risk. Nearly 11,000 plasma proteins were assessed at a mean age of 63 years for seven organ clocks, with childhood data (birthweight, adolescent body mass index (BMI) and education), and serial socioeconomic, environmental exposures and lifestyle information in the NSHD. **a**, Proteomic age gap distributions for 1,803 NSHD participants. Age gaps reflect the participants' estimated age relative to their chronological age—positive values indicate an older biological aging phenotype, while negative values indicate a biologically

younger phenotype. **b**, Left: Hazard ratios (HRs) for death, comparing extreme agers (oldest decile for each age gap distribution) with other participants. The conventional age and seven organs were examined with Cox proportional hazards models controlled for sex and chronological age. The term 'conventional' refers to whole body, all measurable age-related proteins. Right: Kaplan-Meier curves for cumulative incidence of death comparing participants who had multiple organs with extreme aging to those who had none. Images are adapted with permission from ref. 81, medRxiv.

of Health and Development (NSHD). This is a unique, longitudinally assessed cohort of over 1,800 participants, followed since birth in 1946 (ref. 81). With the participants nearly all the same age, the broad distribution of organ aging is noteworthy, as shown in Fig. 3, spanning a range of 10 years for most organs. Moreover, there was a notable relationship between extreme aging (defined as top decile of each distribution) and mortality, which was even more pronounced when three or more organs showed extreme aging (Fig. 3b). The organ clocks were adversely affected by overweight in adolescence, lack of education, smoking and various types of adversity across the life course (such as parental divorce or lack of social support). In contrast, there were several healthspan-protective factors including normal birthweight, higher childhood socioeconomic status, sustained physical activity and moderate alcohol intake.

Cell clocks. Plasma proteomic clocks have been extended to model cell-type-specific age, based on circulating proteins derived from

specific cell types⁸². Measuring >7,000 proteins in ~60,000 individuals, biological age was estimated across ~40 cell types spanning neuronal, immune, glial, endocrine, epithelial and musculoskeletal origins. These signatures predicted incident disease and mortality over 15 years. Amyotrophic lateral sclerosis showed the strongest association with skeletal myocyte aging (hazard ratio = 12.7 for extreme accelerated versus youthful aging). AD was associated with accelerated aging across multiple cell types (Fig. 4). In smokers, extreme respiratory epithelial cell aging increased lung cancer risk by 58% beyond smoking alone and in normoglycemic individuals, myeloid cell aging predicted type 2 diabetes risk. A polycellular aging risk score stratified mortality risk across cohorts and proteomics platforms. Notably, proteomic models were largely stable in the above-mentioned UK NSHD cohort across three time points in 3–7 years apart in a subset of 364 individuals assessed longitudinally. This study establishes a framework for quantifying biological aging at cellular resolution using plasma proteomics,

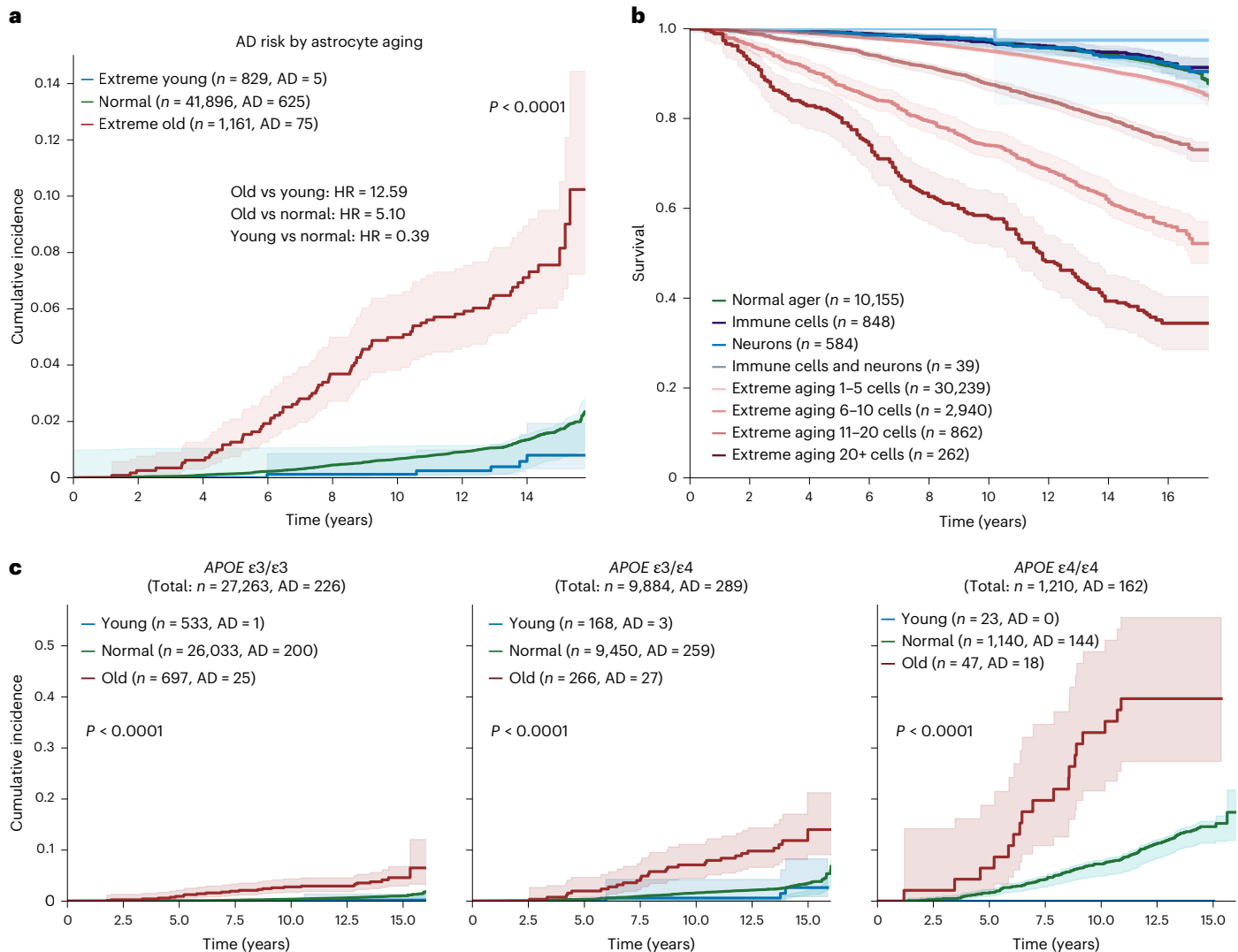


Fig. 4 | Cell-type-specific aging and AD. In all graphs, lines represent Kaplan–Meier estimates of cumulative incidence, and shaded bands indicate 95% confidence intervals. **a**, AD cumulative incidence by astrocyte aging status over 15 years of follow-up in the UK Biobank—extreme agers, normal agers and youthful agers. Overall log-rank test comparing cumulative incidence across extreme young, normal and extreme ager groups. $P < 0.0001$. **b**, All-cause mortality by number of ‘extreme aging’ cell types over 15 years of follow-up in the UK Biobank. Kaplan–Meier curves are stratified by extreme aging burden—normal agers (~90% survival), 1–5 extreme aging cell types (~85% survival),

6–10 (~73% survival), 11–20 (~52% survival) and 20+ (~34% survival). Additional curves show individuals with youthful immune cells or neurons, irrespective of other cell-type aging patterns. **c**, Cumulative incidence of AD by astrocyte aging status (‘extreme old’, ‘normal’ and ‘extreme youthful’ astrocytes) across *APOE* genotypes in the UK Biobank over 15 years of follow-up. Within each *APOE* genotype stratum, group differences were assessed using a two-sided log-rank test comparing the extreme astrocyte aging group with the combined young and normal ager groups. Images are adapted from ref. 82, Springer Nature.

revealing heterogeneous aging trajectories and their impact on disease susceptibility and resilience.

Risk stratification. Organ clocks have been assessed in trials and may have clinical utility for risk stratification. The UK Biobank proteomic data for organ clocks was further validated in biobanks from China and Finland, showing similar prediction accuracy across ancestries and association with 18 chronic diseases⁸³. Individuals with slow clock aging had less than 1% chance of developing AD⁸³. Further validation of organ clocks was seen in a cohort of 1,700 participants with 10-year follow-up. Proteins in the Olink panels were linked with the risk of all-cause mortality and developing 13 diseases, exceeding the prognostic accuracy of electronic medical record data⁸⁴.

Some other recent reports add further texture to the findings about the plasticity of organ clocks. In an extensive sampling of more than 500 human tissues from 76 individuals aged 14 to 66 years, 13 tissue-specific proteomic organ clocks were assessed⁶.

By the age of 50 years, nine of the clocks showed some increased pace of aging, with early changes commonly noted before that age in the aorta.

A human pan-disease blood atlas of over 5,400 plasma proteins in more than 8,200 individuals, including a wellness cohort with longitudinal follow-up of 96 participants, found that most of the proteomic variability was due to disease rather than age, sex or body mass index⁸⁵. Notably, at the individual level, there was a distinct stable protein profile over several years. Specific proteins were found to be candidates for early detection of multiple types of cancer.

Specific proteomic clocks for brain aging have been evaluated. A panel of 25 plasma proteins (from the Somascan proteomic platform) predicted dementia risk over the next 20 years in two different large cohorts⁸⁶. Another study that incorporated brain imaging, Mendelian randomization and the Olink panel of proteins identified a potential causative link for brevicin and the nonlinear pattern of brain aging with proteomic peaks at 57, 70 and 87 years of age¹⁹.

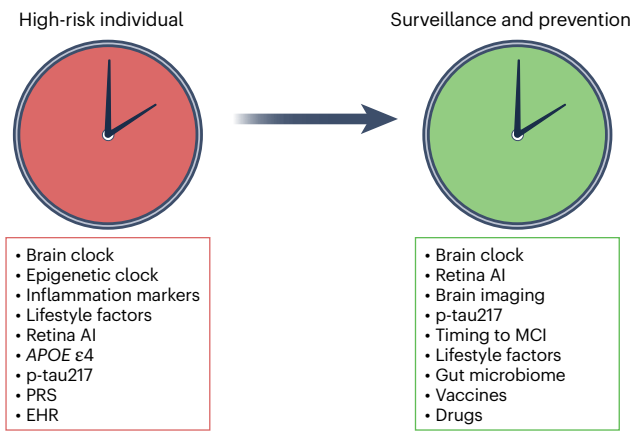


Fig. 5 | A two-stage framework for prevention of neurodegeneration.

High-risk individuals (left) are identified by integrating genetic risk (*APOE* ϵ 4, PRSs), biological-age estimates from epigenetic and proteomic clocks, blood markers (p-tau217, inflammation), retinal AI, lifestyle factors and EHRs. Stratified individuals enter a longitudinal surveillance and prevention program (right) combining repeated biomarker, proteomic clock, imaging, retinal and microbiome measurements to estimate timing to MCI. Preventive interventions span lifestyle modification, geroprotective and disease-modifying drugs, and emerging vaccines, with the goal of delaying or averting progression from preclinical pathology to symptomatic disease. MCI, mild cognitive impairment.

A proteomic healthspan score derived from 53,000 UK Biobank participants with Olink proteins differentiated risk of mortality and risk of more than 17 diseases⁸⁷. The accuracy performance was comparable to epigenetic-based clocks and organ-specific proteomics. As in all the studies that characterized the proteins associated with healthspan and longevity, they were reflective of the hallmarks of aging, particularly affecting the immune response, inflammation, cellular signaling and metabolic regulation.

Other clocks

There are apace of aging clocks based on many other layers of biologic data such as transcriptomics, glycomics, metabolomics, lipidomics, microbiome, telomeric and immune system (discussed in ref. 88). An exhaustive review of transcriptomic clocks in mammals (mouse, rat, macaque and human) found conserved features across all four species, their relationship to lifespan, morbidity and correlation with epigenetic clocks⁸⁹. Likewise, a transcriptomics-based clock known as Transcriptomic Mortality-risk Age used RNA-seq data to predict 4-year all-cause mortality risk⁹⁰ but other omics have been assessed as comprehensively. This is partly because the relatively small number of metabolites and the complexity of the immune system, with its multiple dimensions (cellular, antibody, interferons), does not lend itself to studies at scale in the same way that methylation and proteomic clocks do. Nevertheless, these other omic clocks continue to be actively pursued. A clock using 168 plasma metabolites in more than 225,000 participants was predictive of all-cause mortality, frailty and healthspan⁹¹. A sperm aging clock based on transcriptomic analysis of small noncoding RNAs was recently described and may, in the future, be useful for assessing health risks in offspring in fathers of advanced age⁹².

Integration of multiple omic layers of data, including epigenomics, proteomics, metabolomics and transcriptomics, was the focus of a recent report that identified three distinct clusters of age-related diseases and a common thread of chronic inflammation and immune cell activation⁹³.

The pace of aging of the immune system has been assessed in several small studies, including IMM-AGE, iAge, Immune Health Grade and Immune Health Metric, using a variety of approaches, including T cell receptor and B cell receptor sequencing across the lifespan, single-cell

immune cell analysis, through a panel of 48 cytokines, and multi-omic profiling (single-cell RNA-seq, proteomics, flow cytometry)^{94–101}. It is clear that the interindividual immune response is remarkably variable, as is the extent of immunosenescence and inflammaging with increasing age¹⁰². Undoubtedly, the immune system is not only an important window into the biological aging process, but also a key driver¹⁰³. Its complexity and functionality require evaluation in the context of perturbations such as infection^{103,104}. That has been done with single-cell RNA-seq of immune cell populations after vaccination and infection⁹⁹, and, in a separate study, with multi-omics following the same perturbations, to define immune system dysregulation and its correlation with all-cause mortality¹⁰⁵. A strong case can be made to support the assertion that brain aging is dependent on the fitness of the immune system¹⁰⁶. The interdependence of the brain and immune system has been highlighted¹⁰⁷, which is aligned with the findings of the brain and immune system clocks tracking together, both at the organ and cellular level, and both strongly correlated with outcomes, especially survival, over extended follow-up. This consistent and prominent finding deserves more exploration for the mechanistic underpinnings and to determine whether there is a cause-and-effect relationship.

Not all aging clocks are based on omics technologies. A ‘pace of aging’ clock incorporating standard lab tests (HbA1c, C-reactive protein (CRP) and hemoglobin) with physical parameters (handgrip strength, lung function, balance, gait speed, blood pressure, waist circumference) was used to measure variation in aging rates across 19,000 older adults using longitudinal assessments for over 9 years¹⁰⁸. This clock predicted survival and healthspan, and its performance was comparable to the epigenetic clocks DunedinPACE and GrimAge.

Longitudinal EHRs with laboratory test results have also been used to estimate aging clocks. One such clock (called LifeClock), assessed in nearly 10 million participants, tracked with chronological aging and predicted many age-related diseases accurately¹⁰⁹. Agent-based large language models have used medical records, laboratory tests and lifestyle factors to estimate body-wide and specific organ aging with good performance for predicting all-cause death and several chronic diseases¹¹⁰.

Virtually every type of medical image has been assessed for estimating biological aging, including retinal images, electrocardiograms, magnetic resonance images, bone imaging, CT scans and histology slides^{111–115}. Training AI models with combined CT and fluorodeoxyglucose positron emission tomography imaging can provide data for brain age¹¹⁶. Brain age gaps were also defined by combining MRI and electroencephalogram data¹¹⁷. Adding to the long list data of sources for aging metrics are wearable sensors, facial and skin aging clocks^{88,118}.

Future of aging clocks in the clinic

As reviewed above, many potentially causal links have been established between epigenetic and proteomic clocks and various diseases—especially when these clocks are coupled with genomic data and Mendelian randomization—that help understand the biology of aging and mechanism of diseases^{119–121}. New insights are gleaned, such as proteomic assessment linking cardiometabolic disease and fractures¹²². Many new protein biomarkers have been identified, and some represent strong potential targets for drug development, such as for preventing AD or cardiovascular diseases^{119,123}.

Beyond advancing our understanding of the molecular basis of aging, the clocks can be useful to inform us of the mutability of these aging processes in response to various risk factors or interventions. As mentioned, the eight-decade NSHD study of proteomic organ clocks found evidence for nine protective factors⁸¹, as have epigenetic clocks for lifestyle interventions¹²⁴. Epigenetics and proteomic-based organ aging have been linked to specific medications, lifestyle behaviors, menopause, foods and occupations¹²⁵. The GrimAge and other epigenetic clocks can be favorably shifted by exercise, a finding that was more pronounced with higher levels of physical activity and consistent

across studies^{124,126}. Less impact, but still slowing of epigenetic age, was seen for omega-3 and vitamin D supplementation, multivitamins and the Shingles vaccine^{41,66,67}. The GLP-1 drug class has been associated with slowing the aging process according to epigenetic, proteomic and transcriptomic clocks in rodents, epigenetic clocks in people with obesity or diabetes, and by favorable proteomic profiles from randomized clinical trials^{123,127,128}.

As the quest to develop a medication that slows aging gathers momentum, it is possible that aging clocks could serve as a surrogate metric in clinical trials, especially when they are assessed orthogonally and show high magnitude and consistency of effect, for regulatory approval. This would be a far more practical means of getting an extension of healthy longevity into the clinic than requiring a decade or more of follow-up for major outcomes, but that potential use case for clocks is far from settled.

Looking beyond clinical trial assessments, it is likely that refinement of aging clocks will lead to their mainstream use for clinical care in the coming years, facilitating primary prevention of age-related diseases, particularly cardiovascular, cancer and neurodegenerative conditions. Many developments with AI will ultimately help propel this forward, such as foundation models that can assess the entire methylome instead of a few hundred CpG sites, or a more comprehensive proteome, at scale^{129,130}.

Whereas large language models tokenize words, the first large-scale, health-focused generative AI model Delphi-2M tokenizes health data and events—greatly exceeding expectations with its predictive capacity for more than 1,200 diagnoses over the next 20 years, conditional on each patient's prior history¹³¹. Notably, this model not only led to prediction of a diagnosis or health event, such as a heart attack, but also when it would occur. Surprisingly, that was based on GPT-2, a low-parameter transformer model, and only clinical and lifestyle data from 400,000 UK Biobank participants (replicated in 1.9 million Danish participants). The addition of polygenic risk score (PRS) data improved its predictive accuracy. It would be anticipated that using current models such as GPT-5 or Gemini-3, along with other layers of data, would take the accuracy or prediction to much higher levels.

The practice of medicine today is reactive, after a cancer is detected by symptoms or mass screening, or when a patient develops mild cognitive impairment indicative of AD. With the help of aging clocks and multimodal AI to integrate many layers of data, we have the newfound capability to define high-risk individuals for major age-related diseases. We no longer have to rely on a single layer, such as PRS, or have the constraint of knowing there is risk of a condition but not when it might appear. The pivot to primary prevention of major age-related diseases such as AD (Fig. 5) may well be the singular most important contribution of multimodal AI and aging clocks in the future.

To that end, a clinical trial aimed at preventing AD has been initiated (Clinicaltrials.gov identifier [NCT07646054](#)). This trial will randomize participants at high risk of developing AD (APOE4 allele, family history of AD, high PRS and high plasma levels of phosphorylated tau217 (p-tau217)) to an intensive, continuous lifestyle coaching intervention or to receive educational materials only. The primary endpoint is the reduction of p-tau217 and slower aging of the proteomic brain clock. Prior work suggests that lifestyle factors such as exercise can lower p-tau217 and other neuromarkers (p-tau181, neurofilament, GFAP) but the magnitude of the effect in high-risk individuals¹³², and the contribution of sleep quality and diet, remains uncertain. The reduction of p-tau217 and slower brain clock aging are considered as surrogates for delaying or preventing AD, but ultimate acceptance of such proxies has to be further validated by long-term prospective trials. Encouraging in this light is the link between the Shingles vaccine and reductions in dementia and AD, with a correlation with slowed aging across multiple epigenetic and transcriptomic clocks^{66,133}.

Similar approaches can be designed to prevent cancer and cardiovascular diseases, combining aging clocks with EHRs, AI analysis

of medical images that contain rich information not detected by expert clinicians and AI analysis of lab values that pick up important trends even in the normal range, missed by clinicians and useful biomarkers¹³⁴. The science of aging clocks has already contributed to identifying causal links between markers and disease, biomarkers and potential drug targets, and that work will surely progress. AI for drug discovery has received considerable attention, but the opportunity going forward will be to develop drugs for prevention, rather than treatment. In that light, agentic AI has substantial potential to catalyze discovery of drugs for prevention of age-related disease, as demonstrated by ClockBase, an autonomous agentic system that analyzes millions of publicly available methylation and RNA-seq samples from multiple species with 40 aging clock predictions—thereby identifying previously unknown age-modifying interventions and drug repurposing opportunities¹³⁵.

Challenges and barriers

There are many limitations to using aging clocks in the clinic at this juncture. It remains unclear how useful they are at the individual level, as the evidence base includes mainly cross-sectional population cohorts, with few longitudinal studies. For the most part, we do not understand how the multitude of aging-linked markers exert their supposed effects or whether they have a true causal role¹³⁶. The signal-to-noise ratio is low and there is a need to further deconstruct clock signals to understand what is stochastic compared to what is deterministic or programmed. Training sets—such as the UK Biobank, which has been the most common resource used in clock studies—lack diversity. Clocks tend to overestimate age in the young and underestimate age in the old, with inconsistent correction of age bias.

Aging clocks are not standardized, and commercial products are being sold without regulatory approval. Many are lacking robust clinical validation, calling into question their reliability and generalizability beyond the limited contexts and population groups studied. Most commercially available clock tests are based on methylation markers, but the CpG sites measured vary, and individuals who get multiple tests can have markedly varying results. Some examples of commercially available tests are GlycanAge (measures patterns of IgG glycosylation), [TruDiagnostic](#) (epigenetic clock), [Generation Lab](#) (SystemAge epigenetic clock), [TallyAge](#) (epigenetic clock), [DeepLongevity](#) (epigenetic as well as multi-omic tests), [AgelessRx](#) (iollo metabolomic test) and [Viome](#) (BioAge transcriptomic clock). These are expensive, direct-to-consumer offerings ranging from about US\$300 to US\$500 per test. There is potential for misuse of these tests by insurance companies that refuse or charge higher for policies based on their results.

Conclusions

While there is intense interest and enormous investments in reversing or slowing the human aging process, the most impressive contributions to the science of aging have come from molecular clocks. However, these may someday help to validate emerging and future strategies to modify human aging, such as through epigenetic cellular reprogramming or senolytic drugs. We now know that human aging is not a linear process. The pace of aging is highly heterogeneous between individuals and asynchronous within the organ systems of an individual. It is eminently measurable and modifiable, especially through well-recognized lifestyle factors. But the extent to which it can be modified, and whether that will be clinically relevant, is unresolved. The dual application of clocks in identifying and tracking a person's disease risk over time is alluring and could potentially replace chronological age as the primary clinical risk stratification tool. Indeed, some would suggest that chronological age has diminishing relevance as a clinical risk factor¹³⁷. The incorporation of aging clocks into medical practice along with other relevant layers of data and multimodal AI will foster a transition to primary prevention of age-related diseases in the future, the next great frontier in human health.

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Competing interests

T.W.-C. is a cofounder and scientific advisor of Teal Rise and Vero Biosciences. E.J.T. declares no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Tony Wyss-Coray or Eric J. Topol.

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